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Roll number - 2401110

Subject - ASD&D

Agile Software Development and DevOps Lab

Experiment 6

Aim: To study and Implement Agile Software Project lifecycle using Jira with DevOps Pipeline Integration.

LO mapped: LO1

LO: Apply Agile methodologies and use Jira for Sprint Planning, task tracking and integrating with DevOps workflow.

Theory:

What is Jira?

Jira is an Agile project management and issue tracking tool developed by Atlassian. It helps software engineering teams plan sprints, track tasks, manage bugs, and monitor releases across customizable workflows.

Core Concepts & Hierarchy

- Project: A shared workspace representing a product, application, or team initiative.
- Issue Hierarchy:
    - Epic: A large objective or high-level feature broken down into smaller tasks (e.g., *Core Infrastructure*).
    - Story: An end-user feature written from a user perspective (e.g., *Storefront UI*). ○ Task / Bug: Technical implementations or defects to resolve.
    - Sub-task: Granular steps within a story or task.
- Product Backlog: A prioritized list of all pending features, enhancements, and bug fixes awaiting assignment.
- Sprint & Scrum Board: A time-boxed cycle (typically 1–2 weeks) where team members pull items from the backlog into active workflow columns (*To Do*, *In Progress*, *In Review*, *Done*). ● Story Points: Relative estimates assessing effort, technical complexity, and risk for each work item.

Jira Setup & Deployment Models

- Jira Cloud (SaaS): Hosted directly by Atlassian. No manual server maintenance or database installations required; teams create an organization site (e.g., company.atlassian.net) and get instant access via a web browser.
- Jira Data Center (Self-Managed): Installed on private infrastructure or dedicated cloud servers (AWS/Azure) for enterprises requiring localized data governance, custom security policies, and high-availability clustering.

DevOps Integration & Automation Lifecycle

- Version Control Linkage: Integrates with code repositories (such as GitHub, GitLab, or

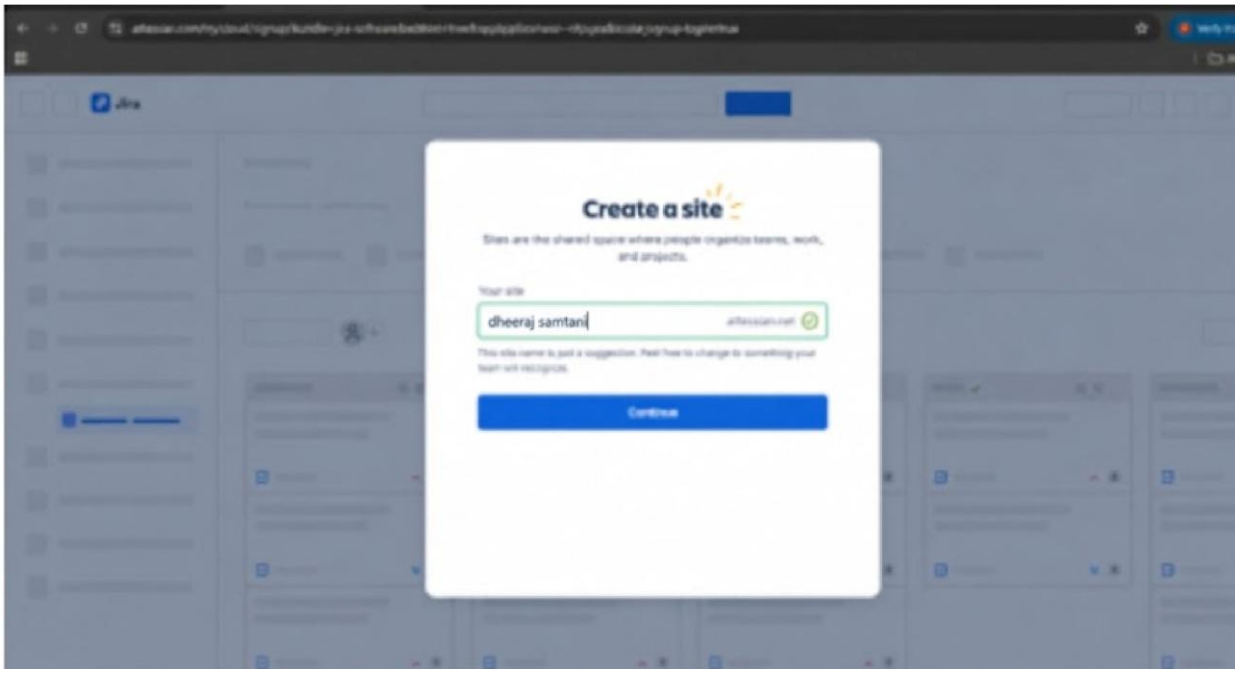
Bitbucket) through official marketplace apps (e.g., *GitHub for Jira*).

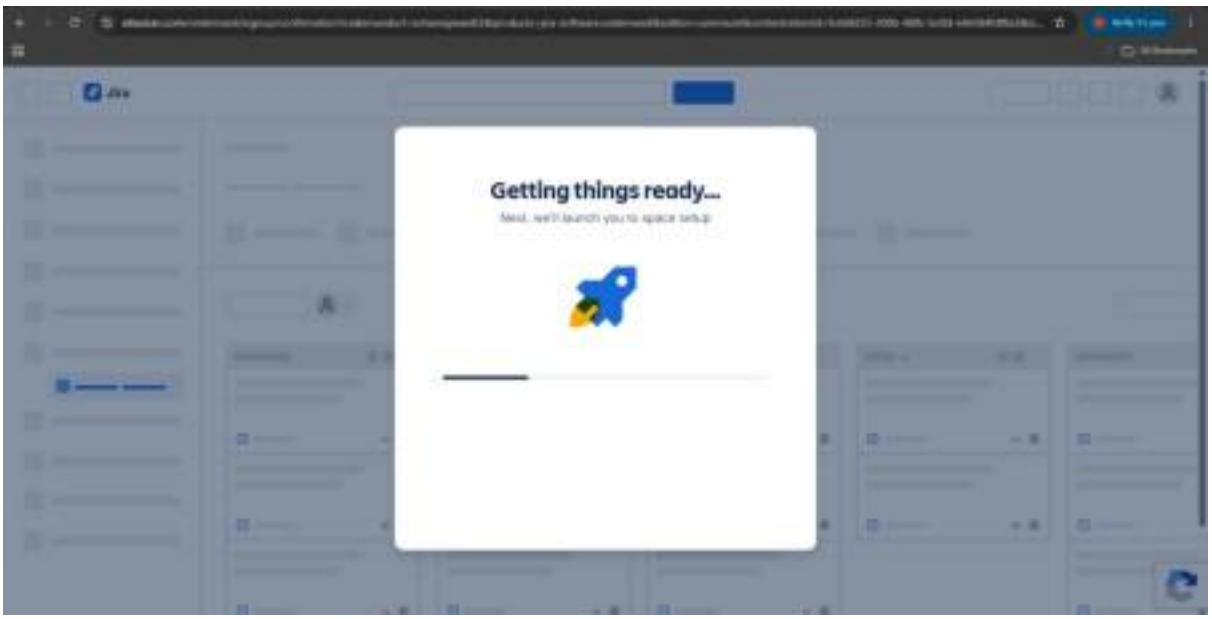
- Smart Issue Keys: Mentioning ticket identifiers (e.g., SCRUM-4, SCRUM-6) in branch names, commit messages, or Pull Request titles binds VCS activity directly to the Jira issue panel.
- Jira Automation Rules: Event-driven rules that trigger state transitions across the board (e.g., automatically shifting a ticket to *In Review* when a Pull Request is opened, or transitioning to *Done* upon a merged PR).

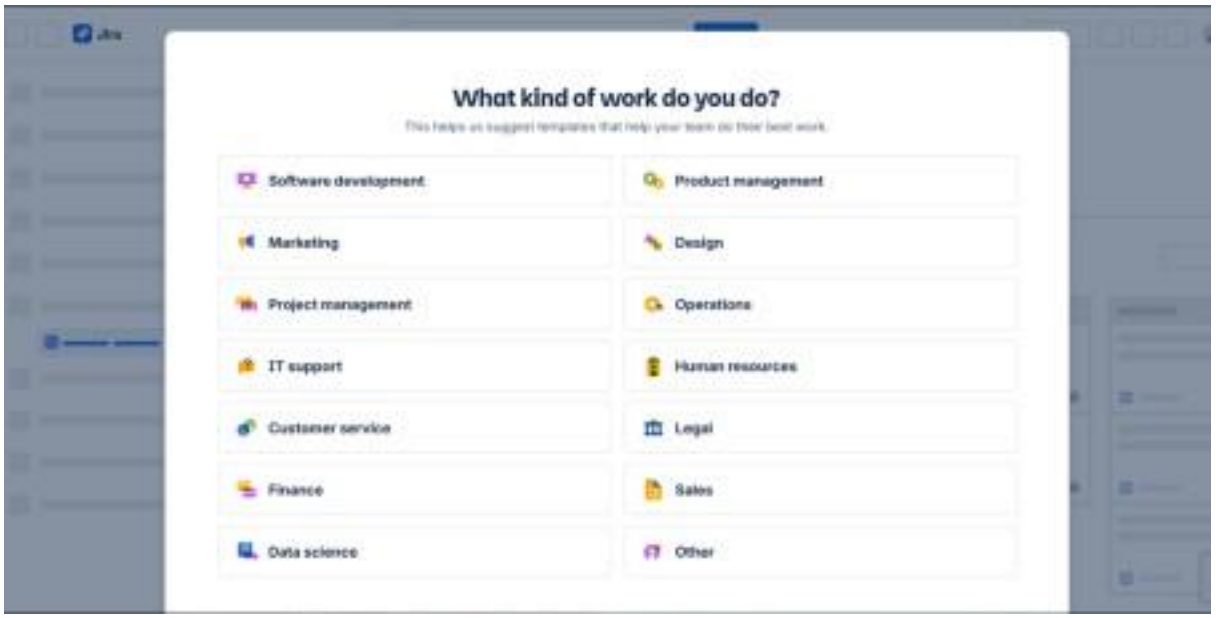
Team Collaboration & Access Management

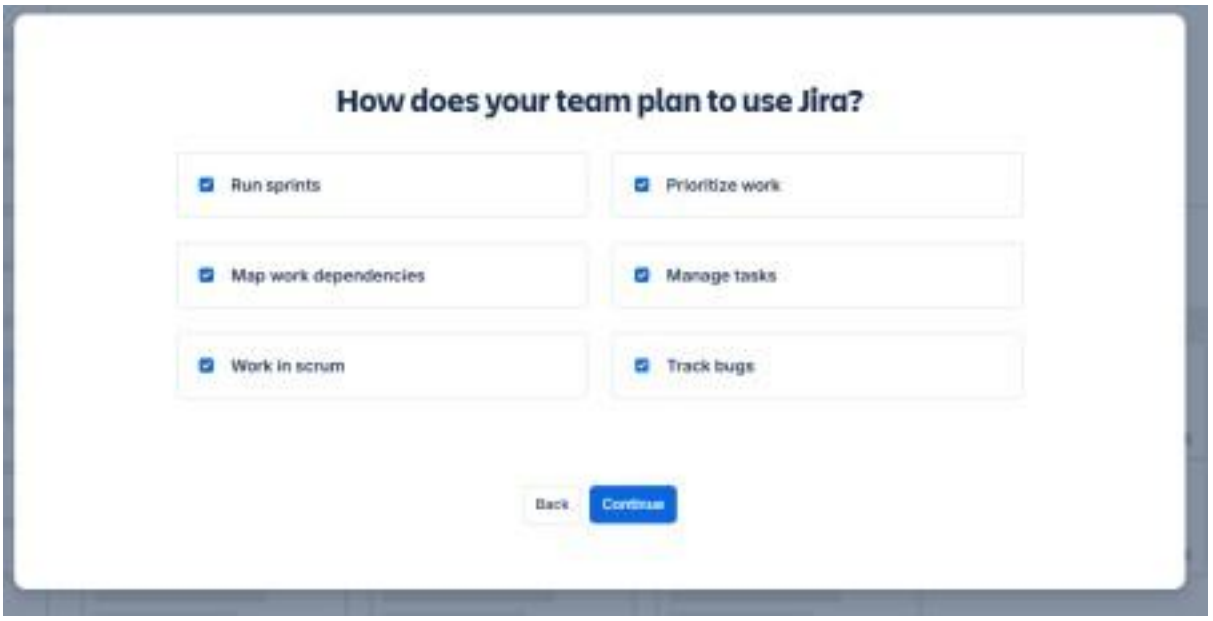
- Role-Based Access Control (RBAC): Access is governed at the organization and project levels via roles (*Administrator*, *Member*, *Viewer*).
- Work Allocation: Tickets are assigned directly to individual developers, allowing visibility into sprint capacity, bottlenecks, and velocity reports.

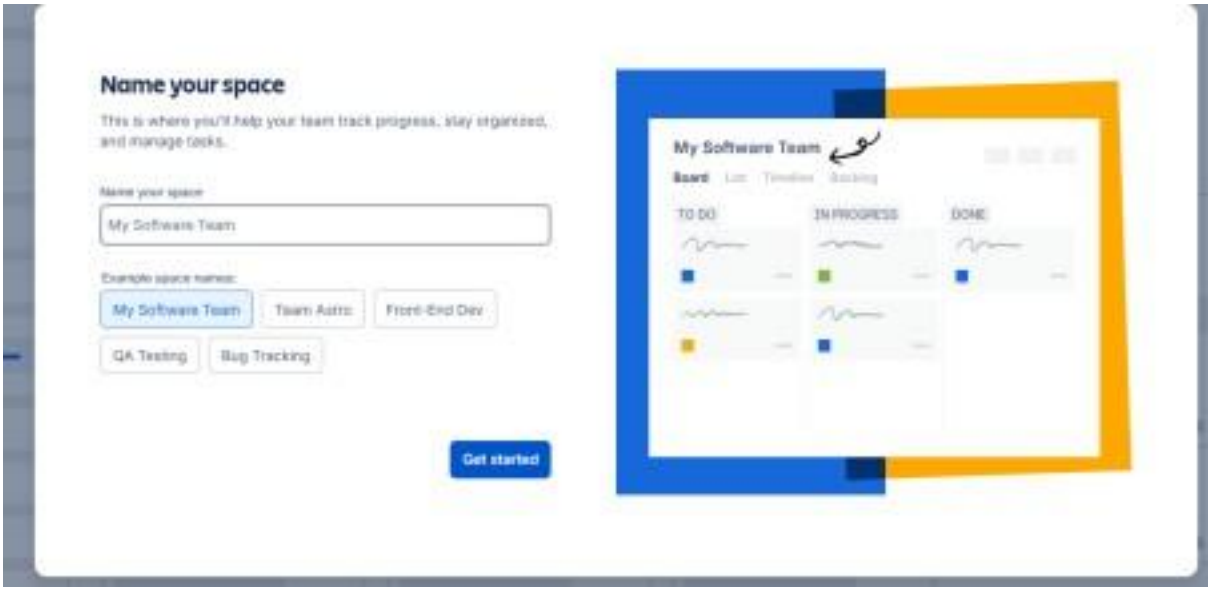
Output:

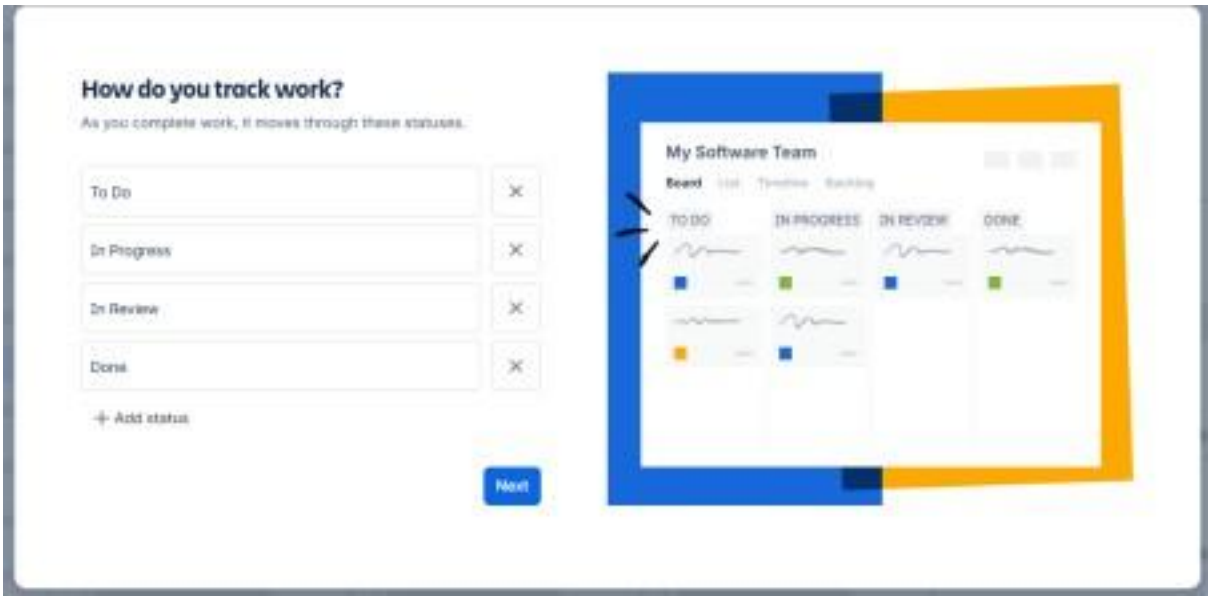





What kind of work do you do?
Software development
Product management
Marketing
Design
Project management
Operations
IT support
Human resources
Customer service
Legal
Finance
Sales
Data science
Other

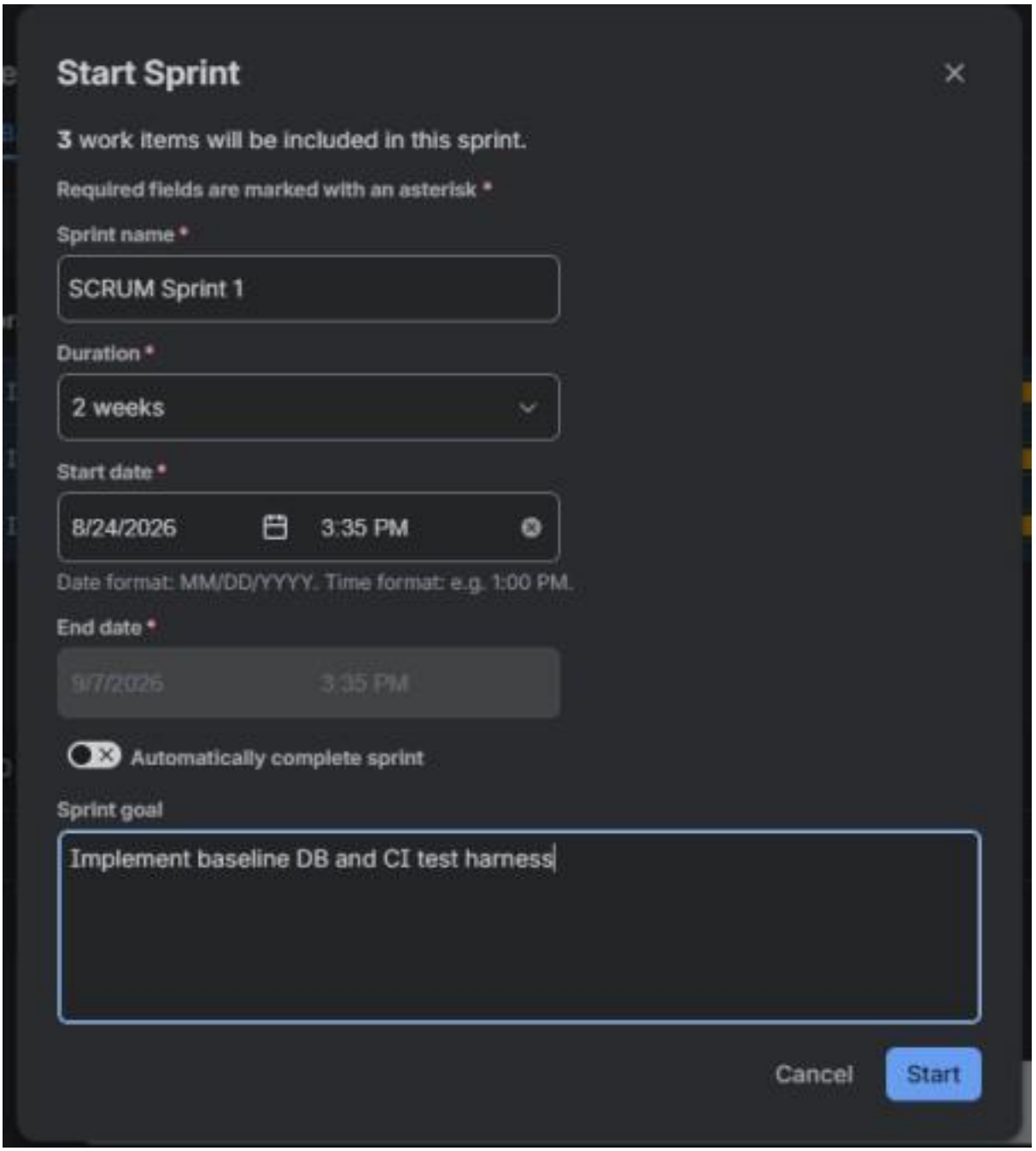

How does your team plan to use Jira?
Run sprints
Prioritize work
Map work dependencies
Manage tasks
Work in scrum
Track bugs
Back
Continue


Name your space
This is where you'll help your team track progress, stay organized, and manage tasks.
Name your space
My Software Team
Example space names:
My Software Team
Team Astro
Front-End Dev
QA Testing
Bug Tracking
Get started
My Software Team
Board
TO DO
IN PROGRESS
DONE


How do you track work?
As you complete work, it moves through these statuses.
To Do
In Progress
In Review
Done
Add status
Next
My Software Team
Board
TO DO
IN PROGRESS
IN REVIEW
DONE


Plan your sprint
Create


SCRUM Sprint 1
Add dates
Start sprint
Core Infrastructure
To Do
Create


Start Sprint
3 work items will be included in this sprint.
Required fields are marked with an asterisk *
Sprint name *
SCRUM Sprint 1
Duration *
2 weeks
Start date *
8/24/2026
3:35 PM
Date format: MM/DD/YYYY. Time format: e.g. 1:00 PM.
End date *
9/7/2026
3:35 PM
Automatically complete sprint
Sprint goal
Implement baseline DB and CI test harness
Cancel
Start


SCRUM Sprint 1
Complete sprint
Create

```
{
  "name": "ecommerce-app",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC"
}


added 60 packages, and audited 69 packages in 4s

7 packages are looking for funding
  run `npm fund` for details

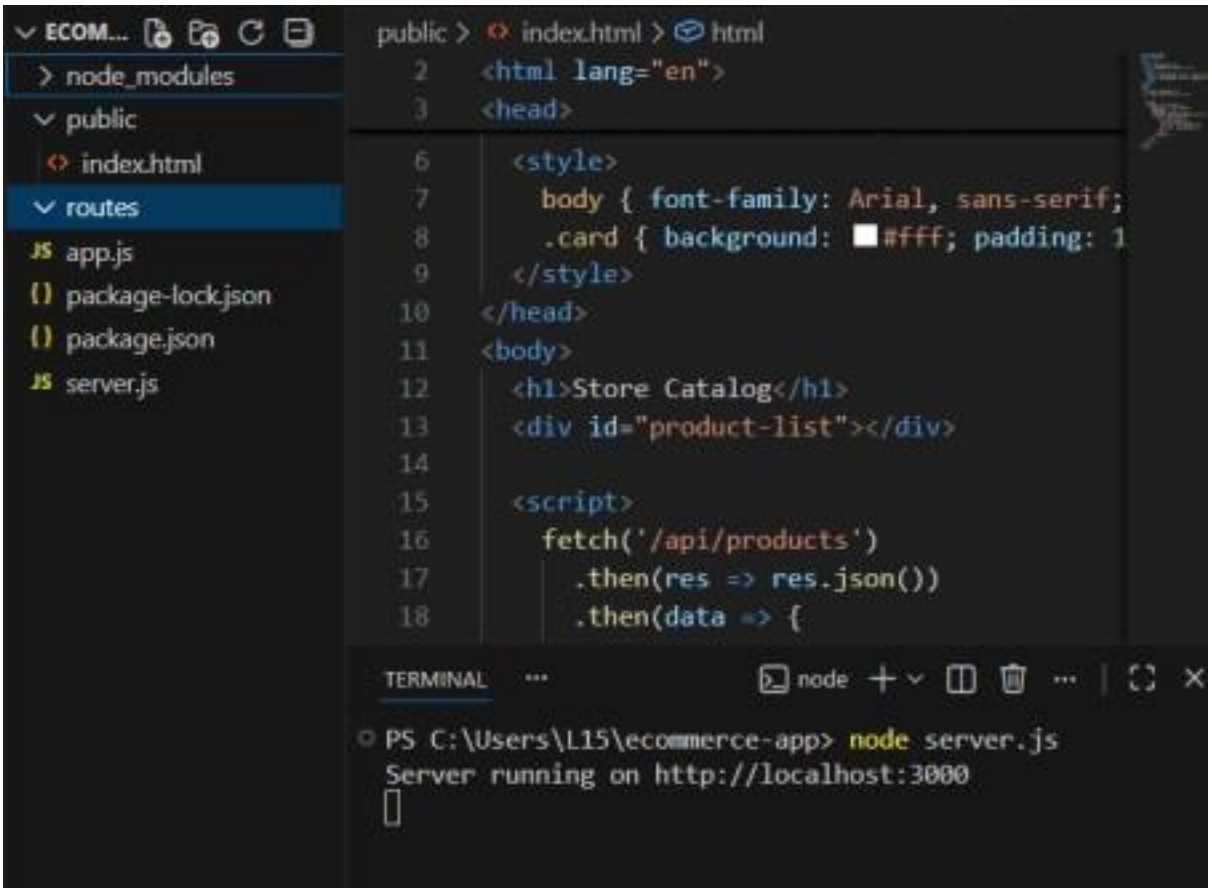
found 0 vulnerabilities

npm WARN deprecated glob@8.1.0: Old versions of glob are not supported, and contain widely published security vulnerabilities, which have been fixed in the current version. Please update. Support for old ver
sions may be purchased at https://npmjs.com/migrate-glob

added 296 packages, and audited 365 packages in 30s

50 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
```



Tech Vault

Cart 0

Premium Gear Catalog

Continuous Integration & Agile Delivered Demo Store



AUDIO

Noise-Cancelling Headphones

$149.99 + Add



PERIPHERALS

RGB Mechanical Keyboard

$99.50 + Add



PERIPHERALS

Wireless Precision Mouse

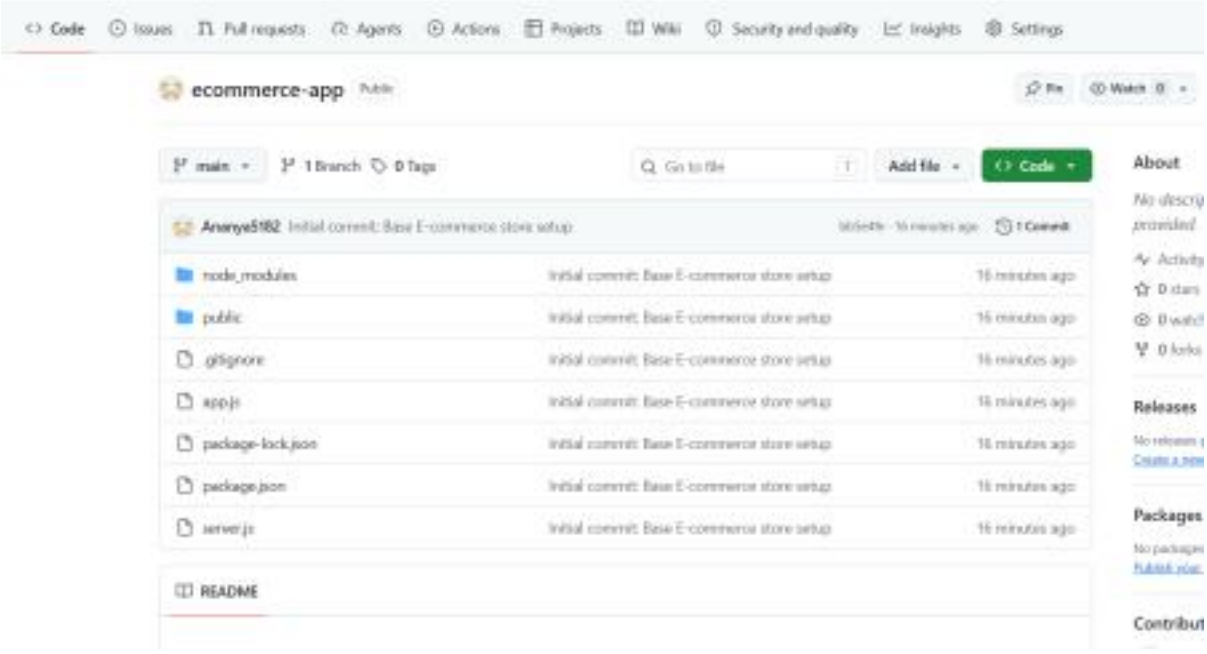
$49.99 + Add

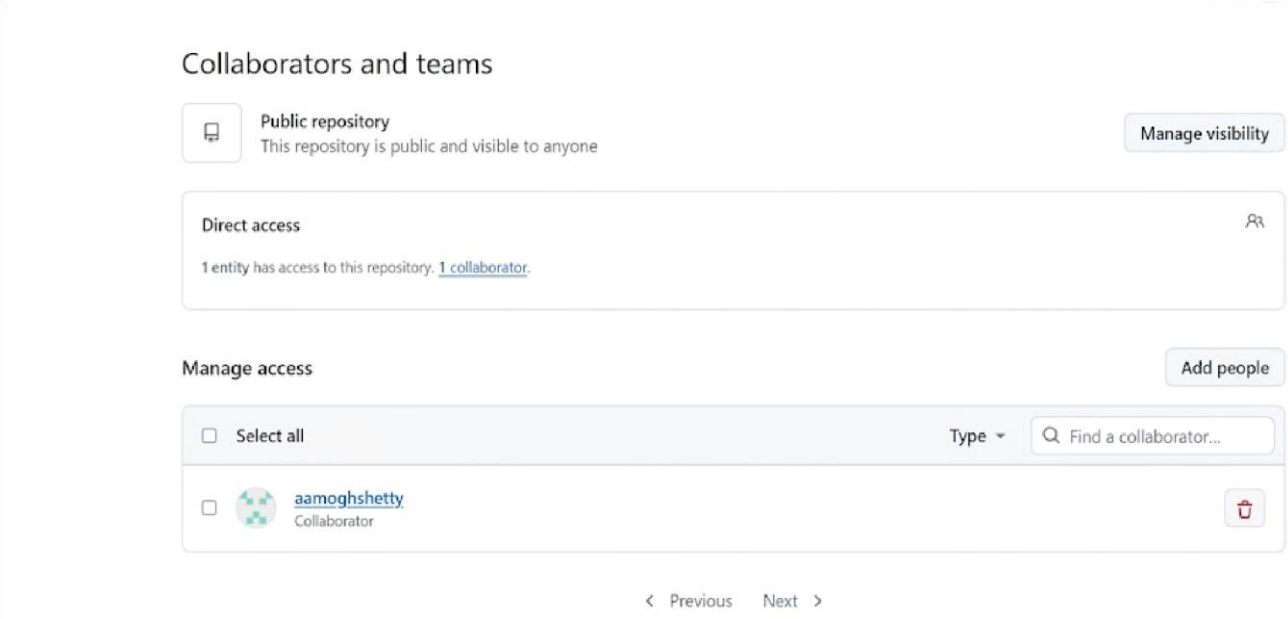


DISPLAYS

4K Ultra-Wide Monitor

$389.00 + Add





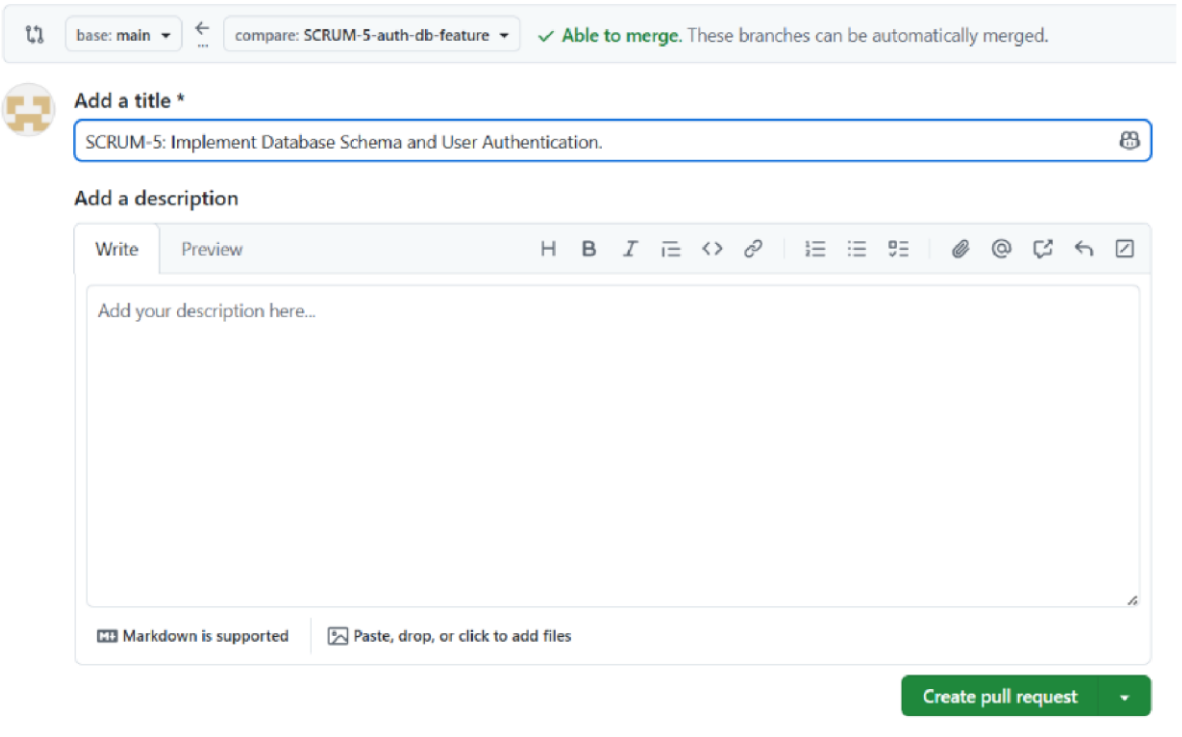

```
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

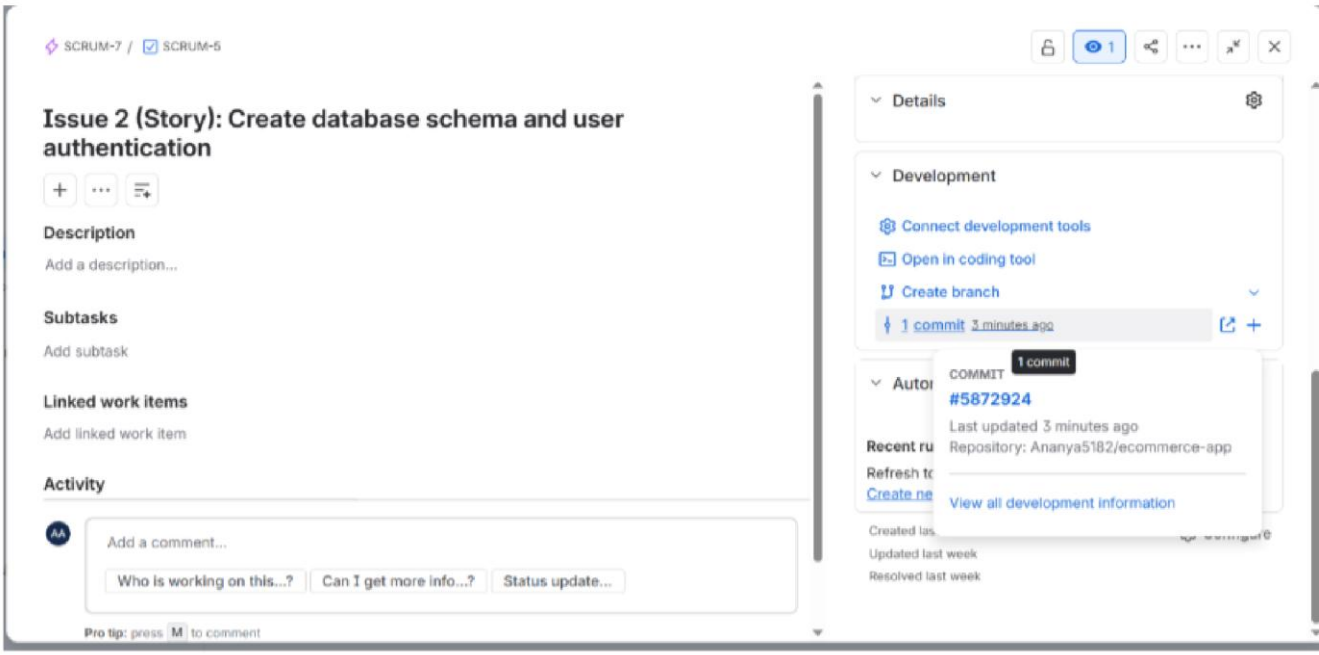
C:\Users\ANGAD\ecommerce-app>git checkout -b SCRUM-5-auth-db-feature
Switched to a new branch 'SCRUM-5-auth-db-feature'

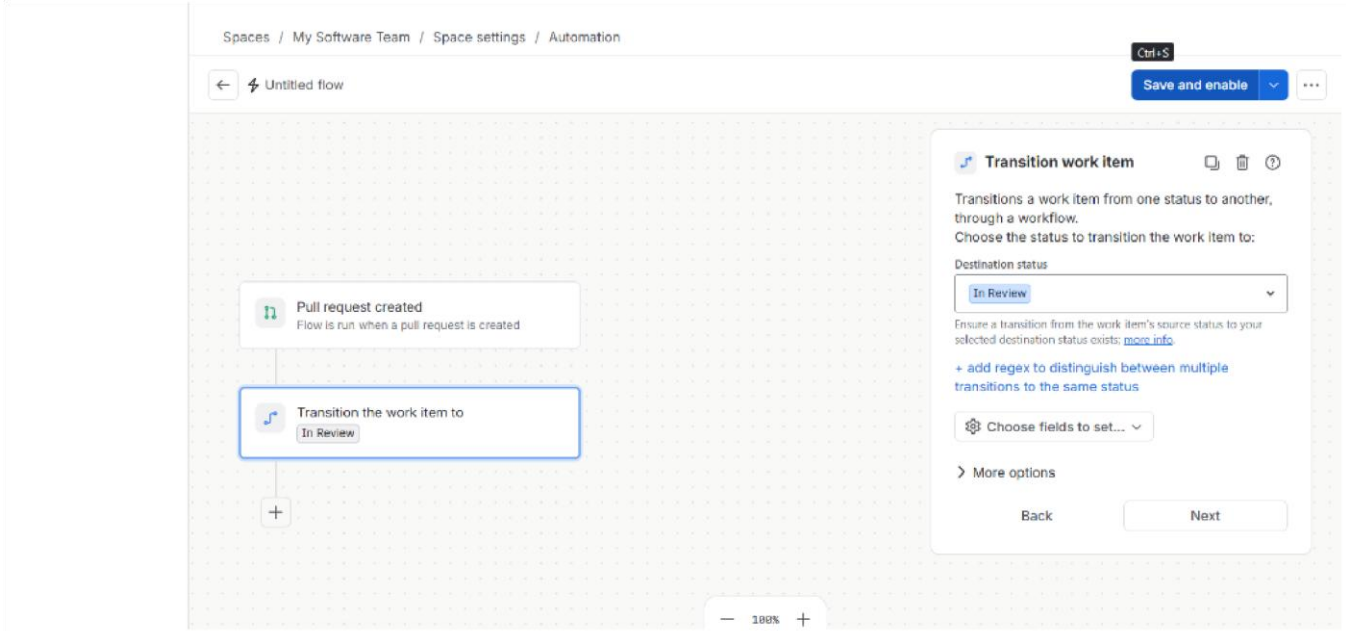
C:\Users\ANGAD\ecommerce-app>git commit --allow-empty -m "SCRUM-5: Connected database schema and user authentication structure"
[SCRUM-5-auth-db-feature 5872924] SCRUM-5: Connected database schema and user authentication structure

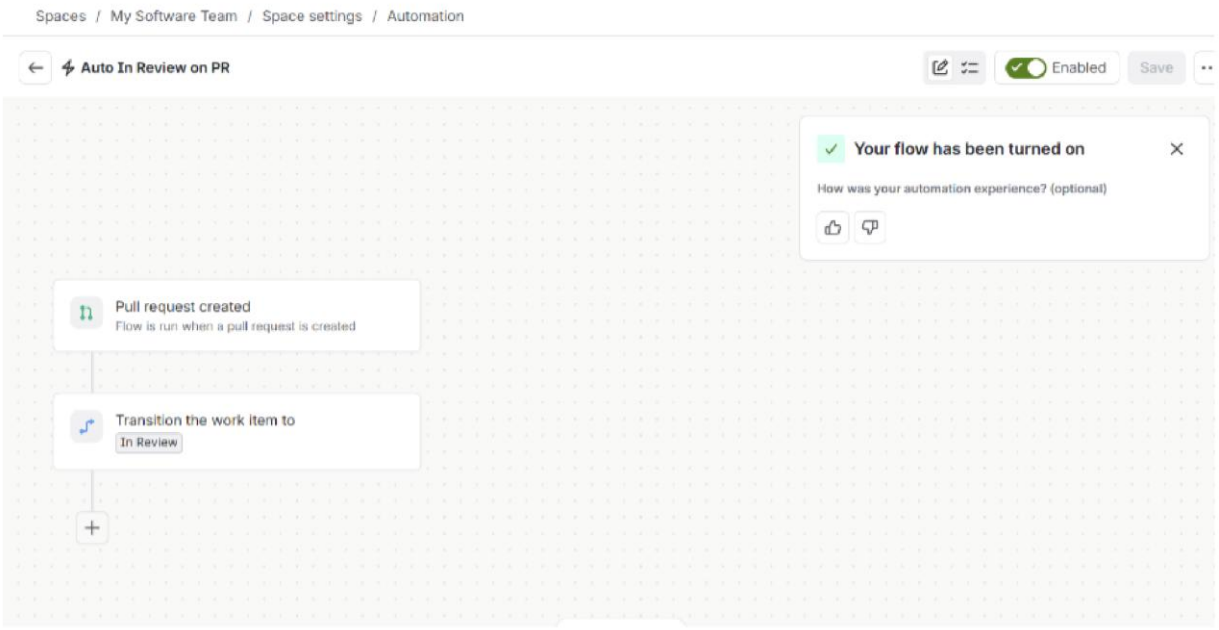
C:\Users\ANGAD\ecommerce-app>git push -u origin SCRUM-5-auth-db-feature
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 229 bytes | 229.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'SCRUM-5-auth-db-feature' on GitHub by visiting:
remote:      https://github.com/Angad5182/ecommerce-app/pull/new/SCRUM-5-auth-db-feature
remote:
To https://github.com/Angad5182/ecommerce-app.git
 * [new branch]       SCRUM-5-auth-db-feature -> SCRUM-5-auth-db-feature
branch 'SCRUM-5-auth-db-feature' set up to track 'origin/SCRUM-5-auth-db-feature'.

C:\Users\ANGAD\ecommerce-app>
```

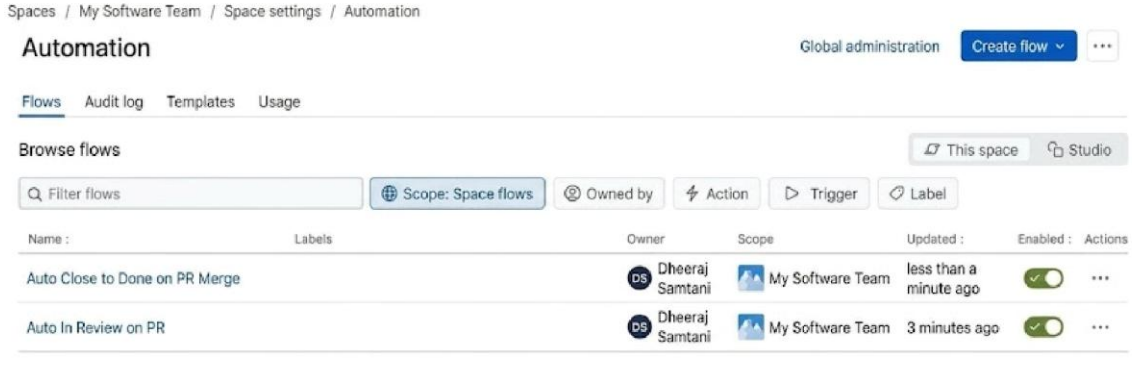

base: main
compare: SCRUM-5-auth-db-feature
Able to merge. These branches can be automatically merged.
Add a title *
SCRUM-5: Implement Database Schema and User Authentication.
Add a description
Write
Preview
Add your description here...
Markdown is supported
Paste, drop, or click to add files
Create pull request

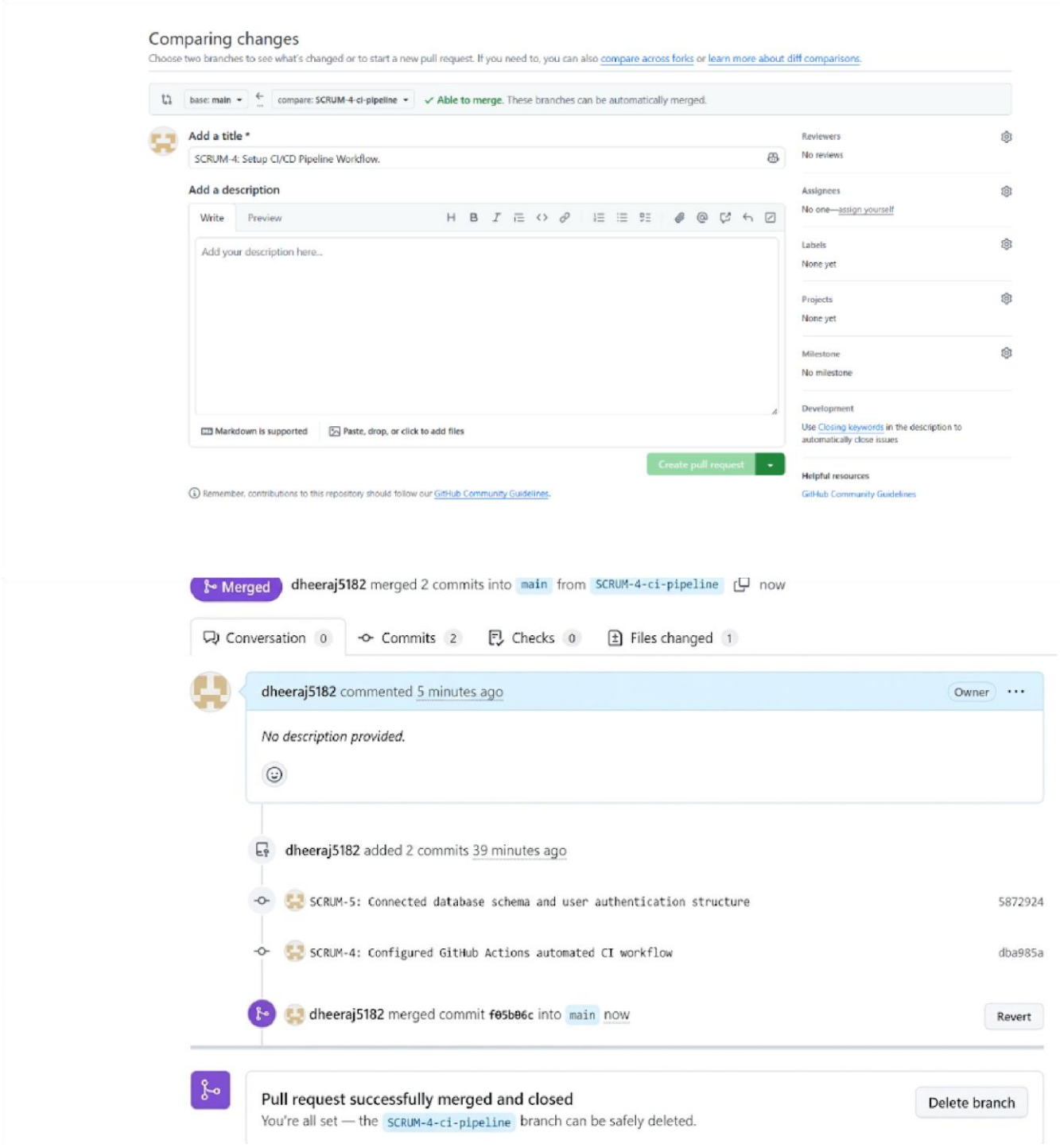

SCRUM-7 / SCRUM-6
Issue 2 (Story): Create database schema and user authentication
Description
Add a description...
Subtasks
Add subtask
Linked work items
Add linked work item
Activity
Add a comment...
Who is working on this...?
Can I get more info...?
Status update...
Pro tip: press M to comment
Details
Development
Connect development tools
Open in coding tool
Create branch
1 commit 3 minutes ago
1 commit
COMMIT
#5872924
Last updated 3 minutes ago
Repository: Ananya5182/ecommerce-app
View all development information
Updated last week
Resolved last week

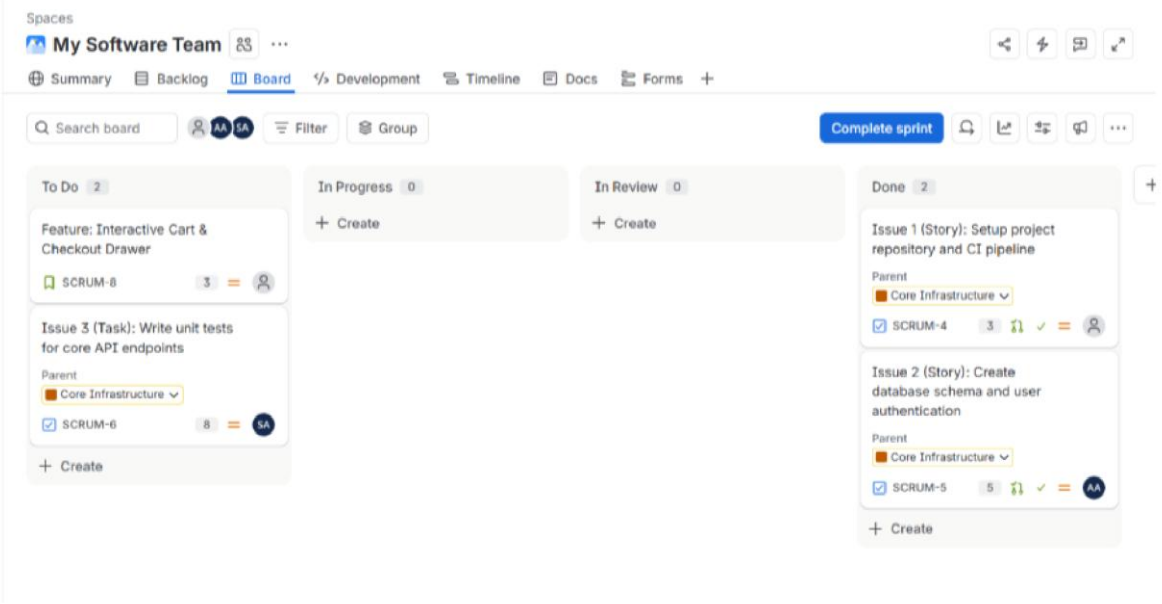

Spaces / My Software Team / Space settings / Automation
Untitled flow
Ctrl+S
Save and enable
Transition work item
Transitions a work item from one status to another, through a workflow.
Choose the status to transition the work item to:
Destination status
In Review
Ensure a transition from the work item's source status to your selected destination status exists: more info.
+ add regex to distinguish between multiple transitions to the same status
Choose fields to set...
More options
Back
Next
Pull request created
Flow is run when a pull request is created
Transition the work item to
In Review
100%


Spaces / My Software Team / Space settings / Automation
Auto In Review on PR
Enabled
Save
Your flow has been turned on
How was your automation experience? (optional)
Pull request created
Flow is run when a pull request is created
Transition the work item to
In Review

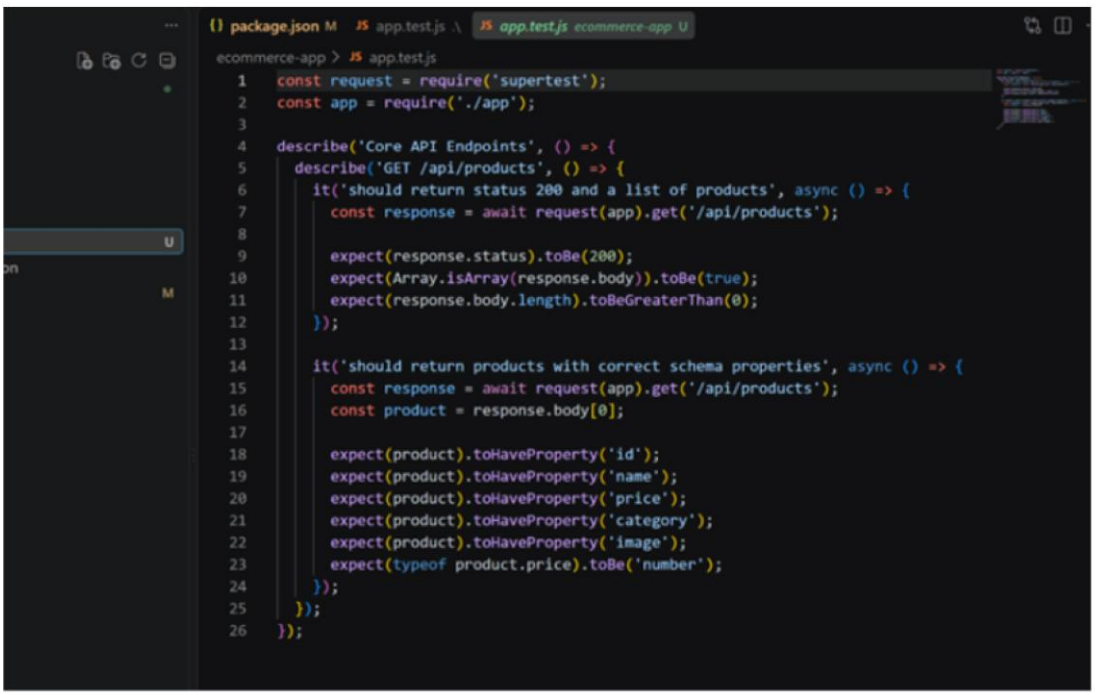

my.software.com/software:/Space settings/Automation


Spaces / My Software Team / Space settings / Automation
Automation
Global administration
Create flow
Flows
Audit log
Templates
Usage
Browse flows
This space
Studio
Filter flows
Scope: Space flows
Owned by
Action
Trigger
Label
Name :
Labels
Owner
Scope
Updated :
Enabled :
Actions
Auto Close to Done on PR Merge
Dheeraj Samtani
My Software Team
less than a minute ago
Auto In Review on PR
Dheeraj Samtani
My Software Team
3 minutes ago


Comparing changes
Choose two branches to see what's changed or to start a new pull request. If you need to, you can also compare across forks or learn more about diff comparisons.
base: main
compare: SCRUM-4-ci-pipeline
Able to merge. These branches can be automatically merged.
Add a title *
SCRUM-4: Setup CI/CD Pipeline Workflow.
Add a description
Write
Preview
Add your description here...
Markdown is supported
Paste, drop, or click to add files
Create pull request
Remember, contributions to this repository should follow our GitHub Community Guidelines.
Reviewers
No reviews
Assignees
No one—assign yourself
Labels
None yet
Projects
None yet
Milestone
No milestone
Development
Use Closing keywords in the description to automatically close issues
Helpful resources
GitHub Community Guidelines
Merged
dheeraj5182 merged 2 commits into main from SCRUM-4-ci-pipeline now
Conversation 0
Commits 2
Checks 0
Files changed 1
dheeraj5182 commented 5 minutes ago
Owner
No description provided.
dheeraj5182 added 2 commits 39 minutes ago
SCRUM-5: Connected database schema and user authentication structure
5872924
SCRUM-4: Configured GitHub Actions automated CI workflow
dba985a
dheeraj5182 merged commit f05b06c into main now
Revert
Pull request successfully merged and closed
You're all set — the SCRUM-4-ci-pipeline branch can be safely deleted.
Delete branch


Spaces
My Software Team
Summary
Backlog
Board
Development
Timeline
Docs
Forms
Search board
Filter
Group
Complete sprint
To Do 2
In Progress 0
In Review 0
Done 2
Feature: Interactive Cart & Checkout Drawer
SCRUM-8
Issue 3 (Task): Write unit tests for core API endpoints
Parent
Core Infrastructure
SCRUM-6
Create
Issue 1 (Story): Setup project repository and CI pipeline
SCRUM-4
Issue 2 (Story): Create database schema and user authentication
SCRUM-5


Development SCRUM-4
Give feedback
Branches
Commits
Pull requests
Builds
Deployments
Feature flags
Other links
Ananya5182/ecommerce-app (GitHub)
Author
ID
Summary
Status
Reviewer
Updated
#2
SCRUM-4: Setup CI/CD Pipeline Workflow.
SCRUM-4-ci-pipeline → main
MERGED
7 minutes ago
We haven't seen any deployments in this space for a while. Finish setting up your deployments with Jira.
Setup Deployments · Dismiss
Description
Subtasks
Activity
Refresh

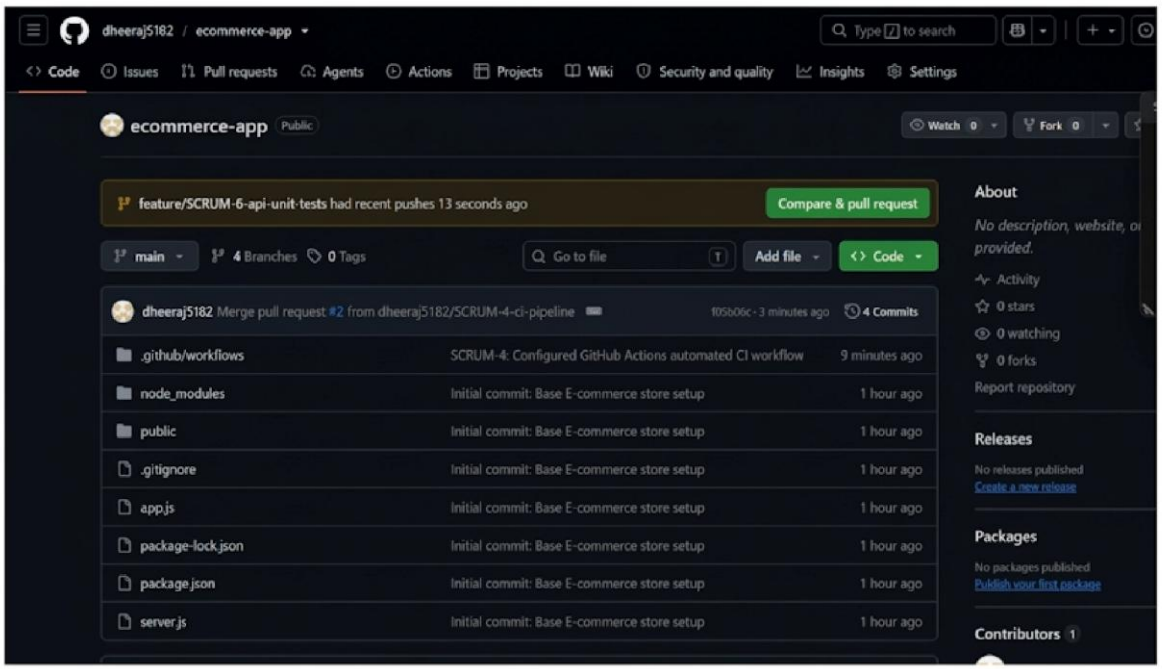

package.json M
app.test.js
app.test.js ecommerce-app U
ecommerce-app > app.test.js
const request = require('supertest');
const app = require('./app');
describe('Core API Endpoints', () => {
describe('GET /api/products', () => {
it('should return status 200 and a list of products', async () => {
const response = await request(app).get('/api/products');
expect(response.status).toBe(200);
expect(Array.isArray(response.body)).toBe(true);
expect(response.body.length).toBeGreaterThan(0);
});
it('should return products with correct schema properties', async () => {
const response = await request(app).get('/api/products');
const product = response.body[0];
expect(product).toHaveProperty('id');
expect(product).toHaveProperty('name');
expect(product).toHaveProperty('price');
expect(product).toHaveProperty('category');
expect(product).toHaveProperty('image');
expect(typeof product.price).toBe('number');
});
});
});

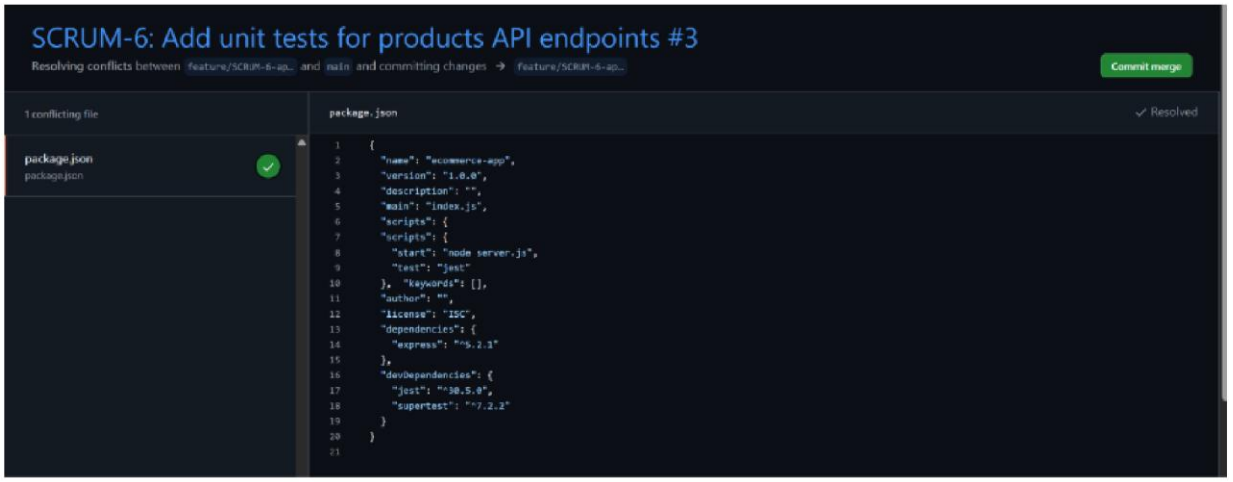
```
PS C:\Users\dheeraj\Documents\ASD-D_Aamogh\ecommerce-app> node
v24.20.0
PS C:\Users\dheeraj\Documents\ASD-D_Aamogh\ecommerce-app> npm
11.19.0
PS C:\Users\dheeraj\Documents\ASD-D_Aamogh\ecommerce-app> npm

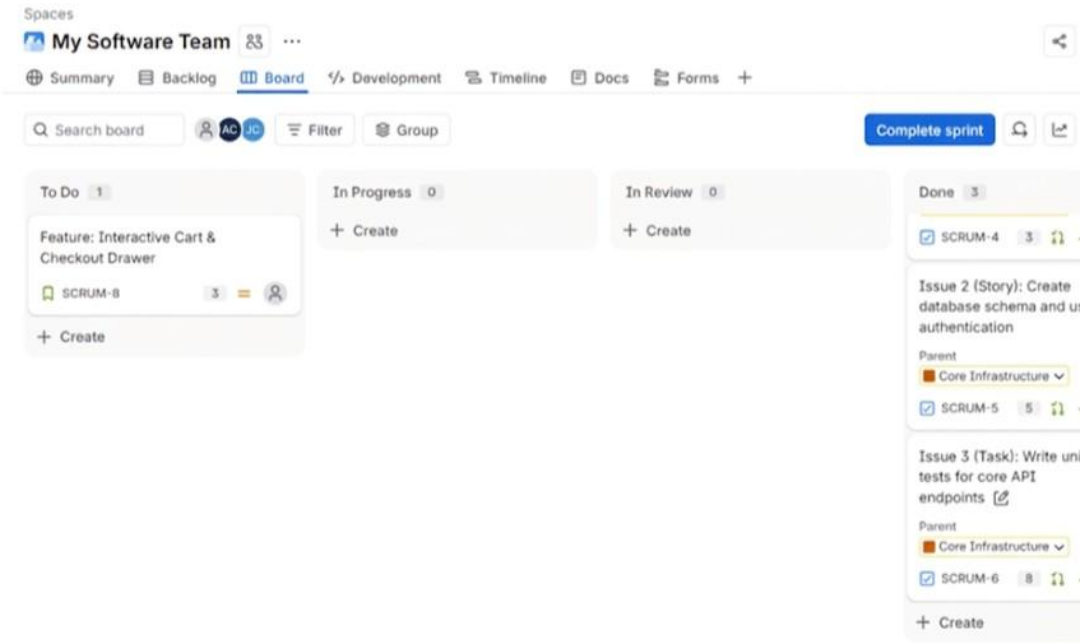
> ecommerce-app@1.0.0 test
> jest

 PASS  ./app.test.js
  Core API Endpoints
    GET /api/products
      ✓ should return status 200 and a list of products (55 ms
      ✓ should return products with correct schema properties

Test Suites: 1 passed, 1 total
Tests:       2 passed, 2 total
Snapshots:   0 total
Time:        1.061 s
Ran all test suites.
npm notice
npm notice New major version of npm available! 11.19.0 -> 12.0
npm notice Changelog: https://github.com/npm/cli/releases/tag/
npm notice To update run: npm install -g npm@12.0.2
npm notice
PS C:\Users\dheeraj\Documents\ASD-D_Aamogh\ecommerce-app>
```







Conclusion

Integrating Jira with GitHub established an automated, end-to-end Agile development lifecycle for our E-commerce application. We learned how to organize backlogs into active sprints, assign tasks across team members, and track real-time progress using Scrum boards. By referencing issue keys in branch names and Pull Requests, we enabled seamless traceability between our codebase and Jira tickets. Furthermore, setting up GitHub Actions CI pipelines and Jira Automation demonstrated how modern

DevOps workflows eliminate manual overhead by automating testing and board transitions upon code merges.